

Name: _____

TOPIC TEST**Functions (Advanced)**

- Time allowed: 45 minutes
- Part A: 10 multiple-choice questions (10 marks)
- Part B: 10 free-response questions (40 marks)
- Total: 50 marks

Part A

10 multiple-choice questions

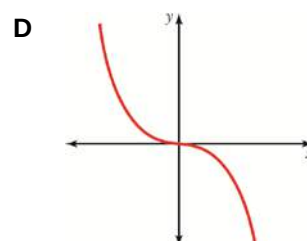
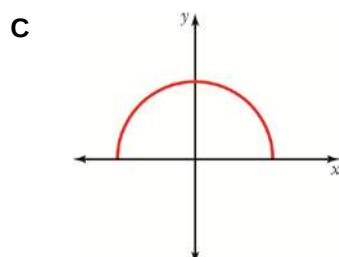
1 mark each: 10 marks

Circle the correct answer.

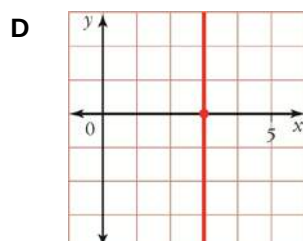
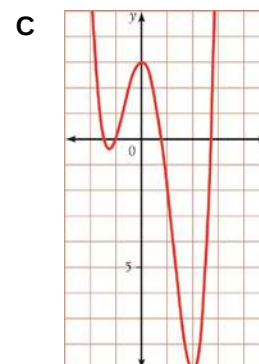
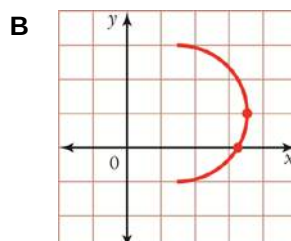
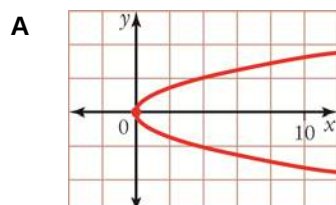
1 Which one of these represents a linear function?

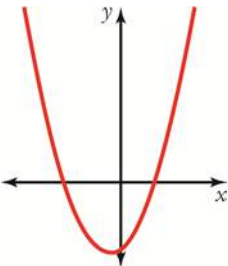
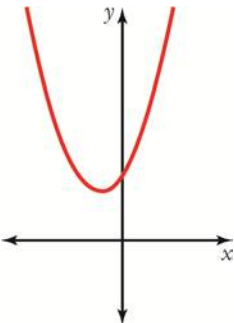
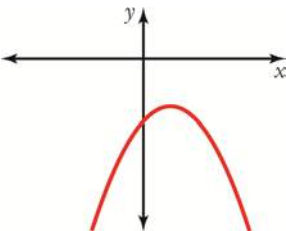
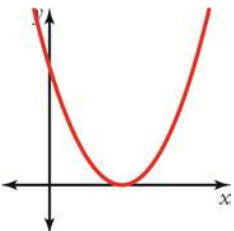
A $(-1, 0), (0, 2), (1, 4), (2, 6)$

B $xy = 1$



2 Which graph represents a function?



- 3 Which statement describes an even function $f(x)$?
- A** Its graph has line symmetry about the x -axis and $f(-x) = f(x)$.
- B** Its graph has line symmetry about the y -axis and $f(-x) = f(x)$.
- C** Its graph has line symmetry about the y -axis and $f(-x) = -f(x)$.
- D** Its graph has rotational symmetry about the origin $(0, 0)$ and $f(-x) = -f(x)$.
- 4 Given $f(x) = \begin{cases} x^2, & x < -2 \\ x+4, & x \geq -2 \end{cases}$, find the value of $f(-2) + f(2)$.
- A** 0 **B** 4
- C** 8 **D** 10
- 5 What is the domain of $y = \sqrt{x+1}$?
- A** $x \geq 0$ **B** $x \geq -1$
- C** $x \geq 1$ **D** $x \leq 0$
- 6 What is the y -intercept of the straight line with equation $2x + y - 6 = 0$?
- A** -6 **B** -2
- C** 2 **D** 6
- 7 Which line is parallel to the line $3x - y + 1 = 0$?
- A** $y = 3x + 1$ **B** $y = 1 - 3x$
- C** $x + 3y + 4 = 0$ **D** $3x - 3y + 1 = 0$
- 8 Find the axis of symmetry of the parabola $y = 2x^2 - x + 1$.
- A** $x = -\frac{1}{2}$ **B** $x = -\frac{1}{4}$
- C** $x = \frac{1}{4}$ **D** $x = \frac{1}{2}$
- 9 Which parabola is the graph of $y = ax^2 + bx + c$, where $a > 0$ and $\Delta = 0$?
- A** 
- B** 
- C** 
- D** 
- 10 The function $f(x) = 2x^2 - x - 6$ has:
- A** 0 real roots **B** 1 real root
- C** 2 real rational roots **D** 2 real irrational roots

Part B

10 free-response questions

40 marks

Show your working where appropriate.

- 11** Find the domain and range of $y = 4 - x^2$.

[2 marks]

- 12** Show that $f(x) = x^3 + 2x$ is an odd function.

[3 marks]

- 13** Find the equation of the straight line passing through $(1, -4)$, that is also perpendicular to the line $y = 2x + 1$.

[4 marks]

14 For the quadratic function $y = 3x^2 + x - 2$:

a state its concavity

b find the value of its x -intercepts

c find the coordinates of its vertex.

[5 marks]

15 Find any values of k for which the quadratic equation $x^2 + 2kx + 9 = 0$ has equal roots.

[4 marks]

16 For the polynomial $P(x) = 2x^6 - x^4 + 3x^3 + x^2 - 4x - 5$,

a state its degree _____

b state the leading term _____

c state the constant term _____

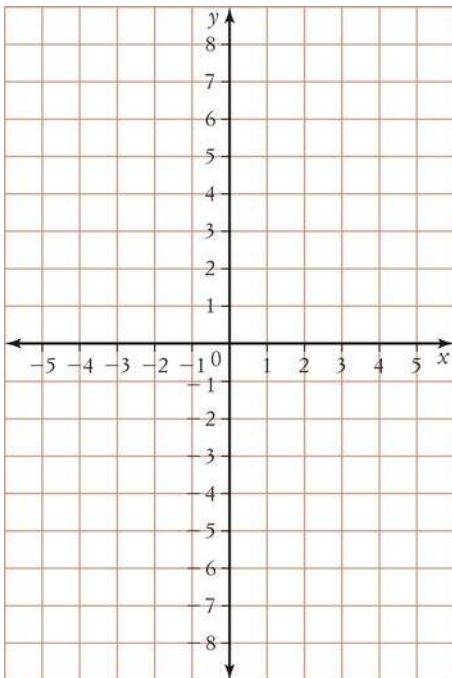
d find the coefficient of x^4 . _____

[4 marks]

- 17 Find any points of intersection between the line $y = x - 3$ and the parabola $y = x^2 + 2x - 3$.

[4 marks]

- 18 Sketch the graph of the cubic function $y = x(x - 3)(x + 2)$, showing the values of all intercepts on the axes.



[4 marks]

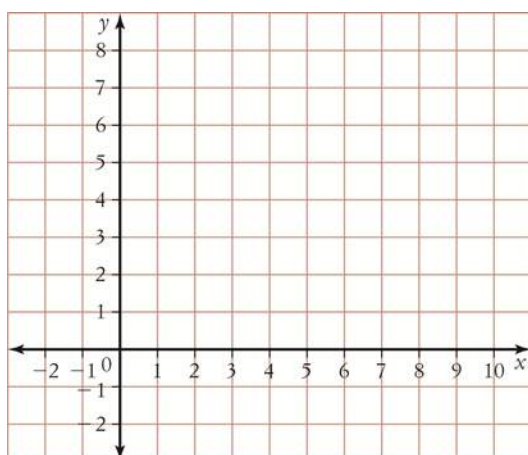
- 19 A cubic function is in the form $y = ax^3 + c$. Find its equation if it passes through (0, 4) and (1, 2).

[4 marks]

- 20 A rectangle has length x and width $5 - x$.

a Write an equation for its area, A .

b Graph the equation on this number plane.



c Find the value of x that gives the maximum area.

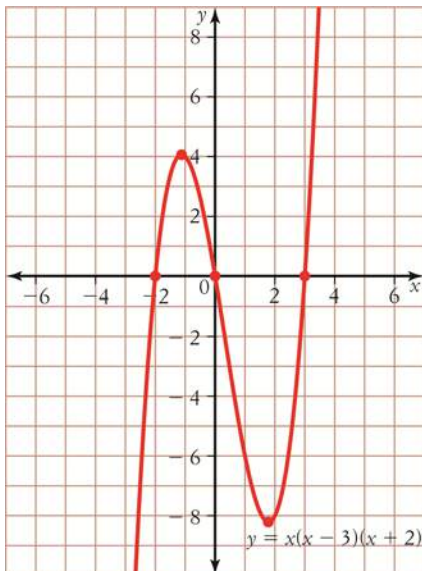
d Find the maximum area of the rectangle.

[6 marks]

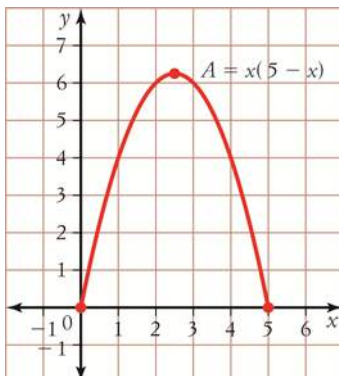
This is the end of the test.

Answers

- 1** A **2** C **3** B **4** C **5** B
6 D **7** A **8** C **9** D **10** C
11 Domain: $(-\infty, \infty)$, Range: $(-\infty, 4]$
12 $f(x) = x^3 + 2x^2$
 $f(-x) = (-x)^3 + 2(-x)^2$
 $= -x^3 + 2x^2$
 $= -(x^3 - 2x^2)$
 $\therefore f(x) = -f(-x)$ is an odd function
13 $x + 2y + 7 = 0$
14 **a** concave up **b** $x = -1, \frac{2}{3}$ **c** $\left(-\frac{1}{6}, \frac{-25}{12}\right)$
15 $k = \pm 3$
16 **a** 6 **b** 2 **c** -5 **d** -1
17 $(-1, -4), (0, -3)$
18



- 19** $y = -2x^3 + 4$
20 **a** $A = x(5 - x)$
b



- c** $x = 2.5$
d 6.25 square units.